Gradual Contract Checking for LLM-Generated Code in a JavaScript Subset

Project Proposal

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Contents

1 Problem Description **1**
1.1 Aims . . . 1

2 Requirements Specification **2**
2.1 Functional Requirements . . . 2
2.1.1 Language . . . 2
2.1.2 Compiler and Runtime Enforcement . . . 2
2.1.3 Static Analysis . . . 3
2.1.4 Evaluation . . . 3
2.2 Non-Functional Requirements . . . 3
2.3 Out of Scope . . . 4

3 Project Plan **4**
3.1 Gantt Chart . . . 4
3.2 Tasks and Milestones . . . 4
3.3 Risk Assessment . . . 5

4 Resources **6**

1 Problem Description

AI-assisted code generation has become a staple of software engineering. In 2026, 83% of respondents to Stack Overflow's Developer Survey used AI tools at work (Stack Overflow, 2026), and GitHub reports that the proportion of pull requests initially drafted by AI more than doubled between March and September 2026 (GitHub, 2026).

However, AI-generated code is often incorrect, insecure, or both. On BaxBench, a benchmark of realistic backend applications, the best-performing model at the time, OpenAI o1, produced correct code for only 62% of tasks, and across models around half of the correct solutions contained security vulnerabilities (Vero et al., 2025).

Trust in LLM output therefore remains a significant factor for AI adoption. Indeed, while 87% of Stack Overflow survey respondents report some level of trust in AI's outputs, 48% trust AI only when they can easily verify its output (Stack Overflow, 2026).

One of the most common methods for developers to carry out this kind of verification for their software is unit testing. Even so, test-passing AI-generated software patches still contain errors at a noticeable rate, with 29.6% of these patches behaving differently from the developers' real fix, and 28.6% of these divergences containing incorrect code (Wang, Y., Pradel, and Liu, 2026).

More fundamentally, tests are a finite set of hand-picked inputs that inevitably miss unconsidered edge cases. As Dijkstra (1970, p. 7) observed, "program testing can be used to show the presence of bugs, but never to show their absence". Dijkstra's own later solution to this problem was to explicitly prove a program's correctness (Dijkstra, 1972).

Hoare (1969) gave this idea a formal basis, and formal verification can now guarantee that a program meets its specification for every possible input. However, these guarantees are expensive. The seL4 microkernel's roughly 8,700 lines of C required around 200,000 lines of Isabelle/HOL proof (Klein et al., 2009), and a survey of 216 practitioners identifies scalability and the expertise required as key barriers to adoption (Gleirscher and Marmsoler, 2020).

Yet the idea at the core of verification, stating explicitly what a program should do and checking the code against that statement mechanically, need not carry this cost. Design by contract (Meyer, 1992) attaches such specifications to each function as preconditions and postconditions, making intended behaviour part of the code rather than leaving it implicit. This suits AI-generated code well: the human states intent as contracts, the LLM writes the implementation, and a checker holds one to the other.

1.1 Aims

Runtime contract systems, such as Findler and Felleisen (2002), share the blind spot of tests, detecting violations only on given inputs. Static verifiers such as Dafny (Leino, 2010) cover every input, but require the programmer to give specifications and annotations until the verifier is satisfied. Gradual verification (Bader, Aldrich, and Tanter, 2018) addresses the weaknesses of each by proving what it can statically and checking the remainder at runtime. For LLM-generated code, existing work primarily uses contracts to guide generation through the prompt, proposing runtime checking that feeds violations back to the LLM as future work (Wang, S., 2026). Execution feedback is known to help LLMs repair their own code (Chen et al., 2024), but whether enforcing contracts helps beyond simply stating them remains an open question.

This project addresses this question. Its aims are to:

- Show that contracts can be enforced gradually in a JavaScript-like language, checked statically where possible and at runtime otherwise.
- Establish how much contract checking can be moved to compile time, and what enforcing the rest costs.

- Find out whether enforcing contracts helps an LLM produce correct code, separating the effect of enforcement from the information the contracts carry.

JavaScript is chosen because it is the most used language in the Stack Overflow Developer Survey, used by 62% of respondents (Stack Overflow, 2026), and LLMs are highly familiar with it, so unfamiliar syntax should not confound the evaluation. The evaluation is scoped to single functions, the granularity at which contracts are written (Meyer, 1992).

If enforcement helps, this supports building contracts into the tools developers use alongside AI coding agents. If not, it suggests that stating constraints matters more than enforcing them, which is useful to researchers designing specification-based workflows for AI-generated code.

2 Requirements Specification

The project will deliver:

- a compiler for the contract language, with a CLI, test suite and static checker
- an evaluation of the static checker and of the LLM experiment, with the harness, task set and contracts used
- the dissertation

Each requirement is given one of three priorities: **must** (essential), **should** (preferable but not essential), or **may** (extension).

2.1 Functional Requirements

2.1.1 Language

- The language **must** support functions, `let`/`const`, conditionals, loops, numbers, booleans, strings, arrays and maps.
- The language **must** support preconditions (`requires`), postconditions (`ensures`) and references to the return value (`result`).
- The language **must** support quantified contracts over arrays, checked at runtime.
- The language **should** support `old()` expressions and loop invariants.

2.1.2 Compiler and Runtime Enforcement

- The compiler **must** transpile programs to readable JavaScript.
- The compiler **must** insert runtime checks for contracts, reporting violations with blame.
- The compiler **must** provide a CLI for compiling and running programs.
- The compiler **must** be covered by an automated test suite.
- The compiler **must** allow the static checker to be disabled, so that every contract is checked at runtime, and allow contracts to be omitted entirely.

2.1.3 Static Analysis

- The analysis **must** be specified precisely, defining the abstract domain, how each construct transforms it and how loops are handled, with a clear argument for why its results are sound, supported by property-based testing.
- The checker **must** classify contracts in loop-free code as proven, refuted or unknown using interval analysis (Møller and Schwartzbach, 2026), which tracks a range of possible values for each numeric variable, starting from the function's preconditions and merging ranges where branches join.
- The checker **should** extend this to loops using widening, narrow ranges using branch conditions, and track array lengths.
- The checker **may** decide relational contracts in loop-free functions by translating each function and its contracts into a logical formula and checking it with the Z3 SMT solver (Moura and Bjørner, 2008), reporting counterexamples for refuted contracts.

2.1.4 Evaluation

- The evaluation **must** use tasks from an existing function-level benchmark, with the benchmark's own tests hidden from the LLM and used to judge correctness.
- The evaluation **must** fix all contracts before any experimental runs, writing them from each task's description and validating them against its reference solution.
- The evaluation **must** measure the proportion of contracts decided at compile time, characterise those that cannot be decided, and measure the runtime overhead with no contract checks, all checks at runtime, and gradual checking.
- The evaluation **must** compare an LLM agent's hidden-test pass rate across three conditions on a pilot set of tasks, with the same model and task prompt in each condition:
  1. *No contracts:* the agent has only the task description.
  2. *Contracts as comments:* task contracts are added to the prompt only as comments.
  3. *Enforced contracts:* task contracts are written in the language and checked by the compiler, with violations reported back to the agent.
- The evaluation **should** extend this comparison to the full task set, with several runs per task.
- If no API budget is available, or the pilot shows a ceiling effect, the evaluation **must** instead measure the proportion of incorrect LLM-generated solutions that contracts detect, how many at compile time, and how many correct solutions are wrongly flagged.

2.2 Non-Functional Requirements

- **Soundness:** a contract classified as proven **must** never fail at runtime, tested with property-based testing.
- **Reproducibility:** model settings and all experimental runs **must** be logged.
- **Cost:** experiments **must** stay within an agreed budget.

2.3 Out of Scope

Classes, asynchronous code, modules, closures, exceptions, mutable objects other than arrays and maps, aliasing, the full standard library, higher-order contracts, multi-function programs, and production-scale systems.

The above requirements are subject to change as the project evolves.

3 Project Plan

3.1 Gantt Chart

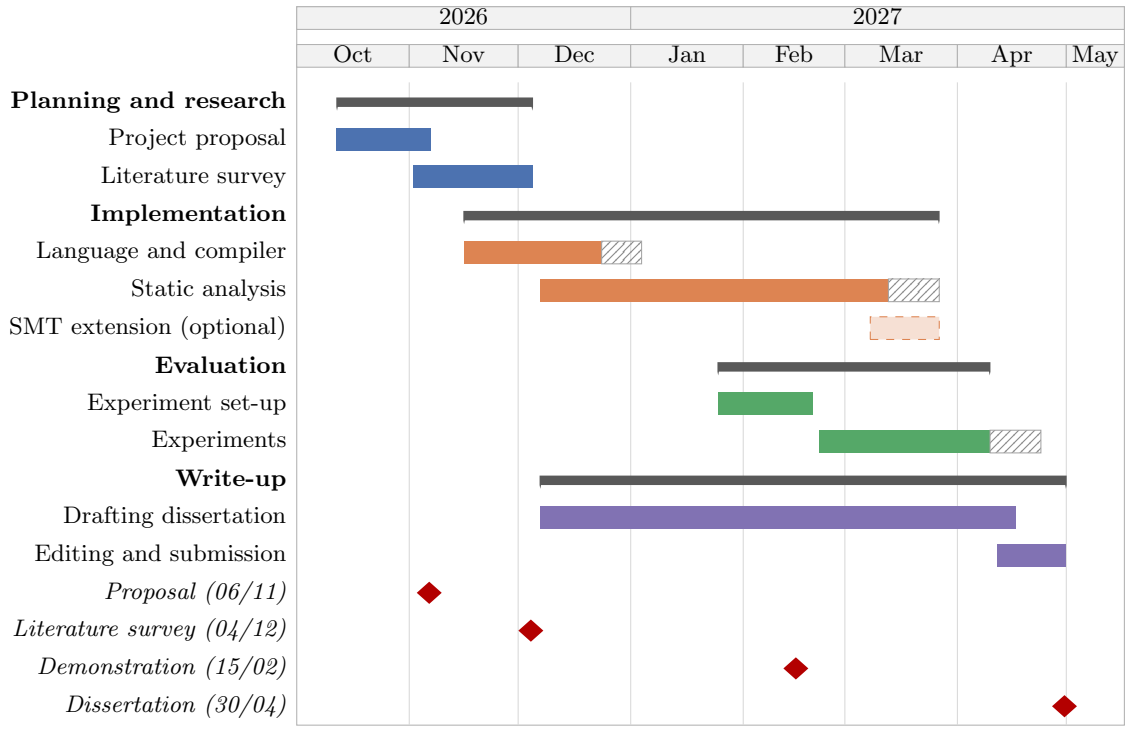


Figure 1: Initial project plan. Diamonds mark submission deadlines. The dashed bar is an optional extension. Hatched segments are contingency time.

3.2 Tasks and Milestones

Table 1 gives the dates and estimated effort for each task in Figure 1, totalling roughly 425 hours excluding the optional SMT extension.

#	Task	Start	End	Buffer	Effort	After	Includes
1	Project proposal	12/10	06/11		25 h	-	problem, requirements, plan
2	Literature survey	02/11	04/12		40 h	1	related work, static analysis reading, benchmark and technology choices
3	Language and compiler	16/11	23/12	03/01	70 h	1	syntax, semantics, parser, JS output, runtime checks, blame, tests
4	Static analysis	07/12	12/03	26/03	115 h	3	interval domain, transfer functions, widening, soundness tests
5	SMT extension (optional)	08/03	26/03		20 h	4	Z3 encoding, loop-free only, counterexamples
6	Experiment set-up	25/01	19/02		30 h	3	task selection, contracts fixed, agent harness
7	Experiments	22/02	09/04	23/04	50 h	4, 6	coverage, overhead, pilot, full run
8	Drafting dissertation	07/12	16/04		70 h	2	chapters written alongside the work
9	Editing and submission	12/04	30/04		25 h	8	revisions, proofreading, submission

Table 1: Tasks, dates and estimated effort. End dates act as waypoints, *Buffer* is the end of each task's contingency time (hatched in Figure 1), and *After* lists the tasks each one builds on, which may still be in progress.

By the progress demonstration on 15/02, the compiler will transpile programs with runtime contract checking and blame (task 3), the static checker will classify contracts in loop-free code (the **must** part of task 4), and the experiment harness will be running on its first benchmark tasks (task 6).

3.3 Risk Assessment

Table 2 lists the main risks to the project, with alternative end-points marked where a risk would change what the project delivers.

Risk	Likelihood	Impact	Mitigation
Static analysis takes longer than planned	Medium	High	Contingency until 26/03, with the SMT extension dropped first *Alternative end-point:* drop loops and evaluate the loop-free checker alone
Other tasks overrun, including through illness	Medium	Medium	Contingency on the compiler and experiments, and Should and May requirements dropped before Musts
No API budget	Medium	High	Open-weight models run on the department's GPU cloud instead *Alternative end-point:* offline detection study on existing LLM solutions
Ceiling effect, where the model already solves nearly every task	Medium	Medium	Detected early by the pilot, then addressed with harder tasks or a weaker model *Alternative end-point:* offline detection study
Static checker is unsound	Low	High	Property-based soundness testing, and runtime checks kept on for proven contracts during the LLM experiments, so unsoundness is detected rather than skewing results
Benchmark unavailable in JavaScript	Medium	Low	Test inputs and outputs translated, which is largely mechanical, with the benchmark chosen during the literature survey

Table 2: Project risks.

4 Resources

Resource	Purpose	Availability
TypeScript and Node.js	Compiler, tests and harness	Free
fast-check	Soundness testing	Free
Z3 (Moura and Bjørner, 2008)	SMT extension	Free
Git and GitHub	Version control and backup	Free
Personal laptop	Development	Available
Commercial LLM API	Agent experiment	Budget requested
Hex GPU cloud	Open-weight models	Department
Benchmark, such as EvalPlus	Tasks and hidden tests	Free
Supervisor	Semantics expertise	Fortnightly meetings
Møller and Schwartzbach (2026)	Interval analysis and widening	Free online
Logic and Semantics unit	Formal semantics and soundness argument	Semester 2

Table 3: Resources, sources of expertise and their availability.

The main limitation is the LLM API budget, which caps the size of the agent experiment. Without it, the fallbacks in Table 2 apply.

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